

Reviewer Pack — Minimal Hodge Span Width Correlation with Resolution Proof W...

Entry fc1a4829f447 · INCONCLUSIVE

SEC autonomous research engine — attribution: Ludovico Kubler

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Minimal Hodge Span Width Correlation with Resolution Proof Width

Entry ID: fc1a4829f447 **Verdict:** INCONCLUSIVE **Recorded:** 2026-06-03 17:59:26 UTC

1. Conjecture

Field A (mathematical branch): Algebraic Geometry (Hodge Theory) **Field B** (complexity object): Resolution Proof Complexity

Statement:

For every CNF formula φ , the minimal Hodge span width ($\text{HSW}(\varphi)$) of its associated algebraic variety is linearly correlated with its resolution proof width $w(\varphi)$, such that $\text{HSW}(\varphi) = \Theta(w(\varphi))$.

Rationale (proposer's reasoning):

Hodge theory provides a rich structure to analyze geometric properties of varieties, and the Hodge span width measures the dimensionality of these spaces. A connection between this geometric invariant and the complexity of finding proofs could reveal new insights into the inherent difficulty of SAT instances.

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): 5cbf1eb7b52d7949

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

The conjecture is supported if the Pearson correlation coefficient between Hodge span width ($\text{HSW}(\varphi)$) and resolution proof width ($w(\varphi)$) for all 30 seeds is greater than or equal to 0.8, with no seed showing a $\text{HSW}(\varphi) > 2 * w(\varphi)$. The conjecture is falsified if any seed yields a correlation coefficient less than 0.8 or a $\text{HSW}(\varphi)$ exceeding $2 * w(\varphi)$.

3. Barrier filter (F1)

Final: PASS

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
RELATIVIZATION	SAFE	0.50	UNCERTAIN	UNCERTAIN
NATURAL_PROOFS	SAFE	0.50	UNCERTAIN	UNCERTAIN
ALGEBRIZATION	SAFE	0.50	UNCERTAIN	UNCERTAIN
KARP_LIPTON	SAFE	0.50	UNCERTAIN	UNCERTAIN

4. Novelty audit

Verdict: NOVEL against 0 hits across arXiv + Semantic Scholar + ECCC

Search queries (3): - "Hodge span width" AND "resolution proof complexity" - "algebraic geometry" AND "proof width" AND "CNF formula" - "minimal Hodge span width" IN title AND "resolution proof width" IN title

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)

Sandbox: isolated subprocess, stdlib-only, ≤90s wall time per cycle **Execution:** rc=1, elapsed=0.1s

5.1 Generated Python source

```
import random
import math
from fractions import Fraction

def gaussian_elimination(A):
    n = len(A)
    for i in range(n):
        # Find pivot in column i
        max_row = i
        for j in range(i+1, n):
            if abs(A[j][i]) > abs(A[max_row][i]):
                max_row = j
        A[i], A[max_row] = A[max_row], A[i]

        # Eliminate entries below pivot
        for j in range(i+1, n):
            factor = Fraction(A[j][i], A[i][i])
            for k in range(n):
                A[j][k] -= factor * A[i][k]
    return A

def hodge_span_width(A):
    rank = 0
    for row in gaussian_elimination(A):
        if any(row):
            rank += 1
    return rank

def resolution_proof_width(phi):
    # Placeholder function to compute the width of a resolution proof
    # This is a dummy implementation and should be replaced with actual logic
    return len(phi) // 2
```

```

def run_trial(seed: int) -> dict:
    random.seed(seed)

    n = random.choice([5, 10, 15, 20, 30, 40])
    phi = [random.randint(0, 1) for _ in range(n)]

    A = [[phi[j] * phi[i] for i in range(n)] for j in range(n)]

    hsw = hodge_span_width(A)
    w_phi = resolution_proof_width(phi)

    if hsw > 2 * w_phi:
        return {
            "metric_name": "Hodge Span Width",
            "metric_value": hsw,
            "instances_tested": 1,
            "n_max": n,
            "conjecture_holds": False,
            "counterexample": f"HSW({phi}) > 2 * w({phi})"
        }

    return {
        "metric_name": "Hodge Span Width",
        "metric_value": hsw,
        "instances_tested": 1,
        "n_max": n,
        "conjecture_holds": True,
        "counterexample": ""
    }

if __name__ == "__main__":
    import sys
    seeds = [int(s) for s in sys.argv[1:]] or [2, 3, 5, 7, 11, 13, 17, 19, 23, 29] * 3

    results = []
    for seed in seeds:
        result = run_trial(seed)
        print(f"TRIAL: {result}")
        results.append(result)

    metric_values = [r["metric_value"] for r in results if r["conjecture_holds"]]
    support_fraction = sum(r["conjecture_holds"] for r in results) / len(results)

    if all(r["conjecture_holds"] for r in results):
        print(f"RESULT: SUPPORTED mean={sum(metric_values)/len(metric_values):.2f} std=0.00 support={support_fraction:.2f}")
    elif any(not r["conjecture_holds"] for r in results):
        first_failing_seed = next(r["seed"] for r in results if not r["conjecture_holds"])
        print(f"RESULT: FALSIFIED counterexample=\"HSW > 2 * w\" first_failing_seed={first_failing_seed}")
    else:
        print("RESULT: INCONCLUSIVE mapping_undefined")

```

6. Per-seed results

(no seed-level data — test crashed before producing TRIAL: lines)

7. Test stdout (last 2KB)

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_d3cf18ca.py", line 83, in <module>
    result = run_trial(seed)
            ^^^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_d3cf18ca.py", line 55, in run_trial
    hsw = hodge_span_width(A)
            ^^^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_d3cf18ca.py", line 37, in hodge_span_width
    for row in gaussian_elimination(A):
               ^^^^^^^^^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_d3cf18ca.py", line 30, in gaussian_elimination
    factor = Fraction(A[j][i], A[i][i])
               ^^^^^^^^^^^^^^^^^^^^^
```

```
File "/usr/lib/python3.12/fractions.py", line 281, in __new__
    raise ZeroDivisionError('Fraction(%s, 0)' % numerator)
```

ZeroDivisionError: Fraction(0, 0)

8. Critic adversarial review

Critic verdict: CHALLENGE

Critic reasoning:

skipped: test crashed before producing data

9. Final verdict & safety rail

Verdict: INCONCLUSIVE

Reasoning:

The test code crashed during execution, which prevents us from verifying the conjecture's conditions. | next: Investigate and fix the crash in the test code to proceed with the verification of the conjecture.

11. Audit log (LLM calls)

Total LLM calls: 11

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
1	propose	ollama_remote	glm4:latest	0	0	18916
2	propose	ollama_remote	glm4:latest	0	0	21571
3	propose	ollama_remote	glm4:latest	0	0	27717
4	preregistration	ollama_remote	glm4:latest	0	0	18340
5	novelty	ollama_remote	glm4:latest	0	0	13929
6	novelty	ollama_remote	glm4:latest	0	0	9082
7	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	15546
8	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	11422
9	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	12055
10	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	9221
11	judge	ollama_remote	glm4:latest	0	0	11433

Totals: 0 input tokens, 0 output tokens, ~\$0.0000 cost, 169231 ms total latency. Provider mix: {'ollama_remote': 11}

(full prompt+response transcripts available in research/audit/fc1a4829f447.jsonl)

12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in research/audit/fc1a4829f447.jsonl for verification of provenance
4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

Sandbox file: research/pvsnp_sandbox/test_*.py (per-cycle)

Replay tarball: research/replay/fc1a4829f447.tar.gz (if generated)